Quantum teleportation

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Move the state,
not the
qubit

Alice and Bob

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- The specific roles they play must be clarified in different situations.
- Additional characters (e.g., Charlie, Diane, Eve, and Mallory) may be introduced as needed.

Remarks on entanglement

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$|\Psi_{AB}\rangle = |\Psi_A\rangle \otimes |\Psi_B\rangle$ → NOT Entangled

$|\Psi_{AB}\rangle \neq |\Psi_A\rangle \otimes |\Psi_B\rangle$ → Entangled ✓

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It is typical in the study of quantum information and computation that we view entanglement as a *resource* that can be used to accomplish different tasks.

When we do this we view the state $|\phi^+\rangle$ as representing one unit of entanglement called an *e-bit.*

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Terminology

To say that Alice and Bob *share an e-bit* means that Alice has a qubit A, Bob has a qubit B, and together the pair (A, B) is in the state $|\phi^+\rangle$.

Teleportation set-up

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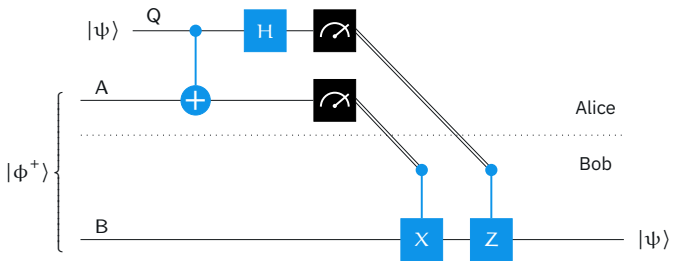
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- The state of Q is “unknown” to both Alice and Bob.
- Correlations (including entanglement) between Q and other systems must be preserved by the transmission.
- The *no-cloning theorem* implies that if Bob receives the transmission, Alice must no longer have the qubit in its original state.

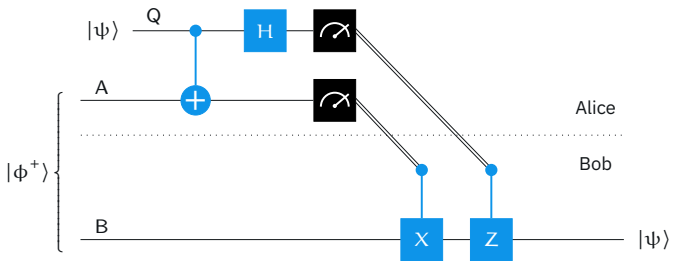
Teleportation protocol

Circuit description



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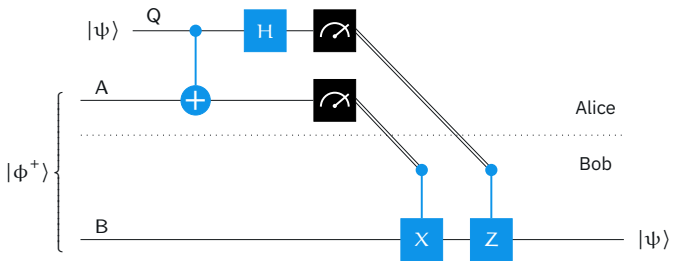


Initial conditions

Alice and Bob share one e-bit: Alice has a qubit A, Bob has a qubit B, and (A, B) is in the state $|\Phi^+\rangle$.

Teleportation protocol

Circuit description

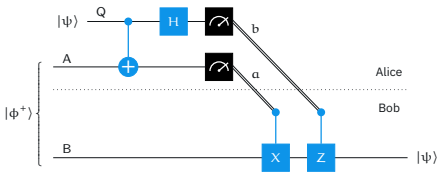


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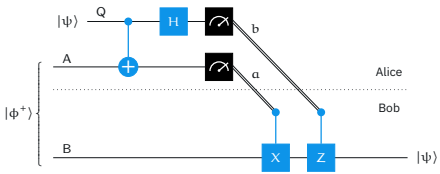
Teleportation protocol



Protocol

1. Alice performs a controlled-NOT operation, where Q is the control and A is the target.

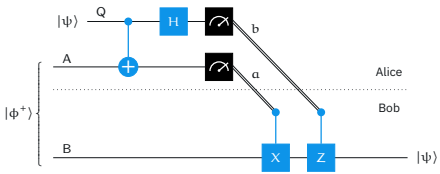
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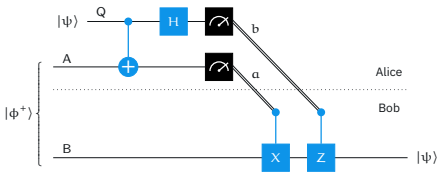
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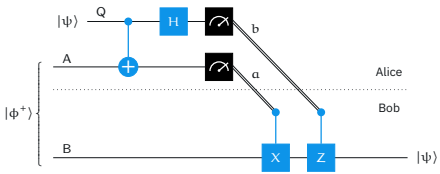
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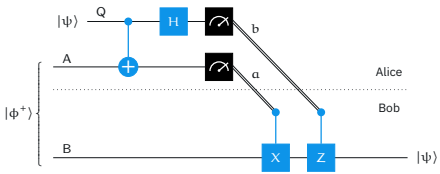
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5. Bob performs these two steps:

   5.1 If $a = 1$, then Bob applies an X operation to the qubit B.

   5.2 If $b = 1$, then Bob applies a Z operation to the qubit B.

Teleportation protocol



Operation performed by Bob

$\mathbb{1}$	if $ab = 00$
Z	if $ab = 01$
X	if $ab = 10$
ZX	if $ab = 11$

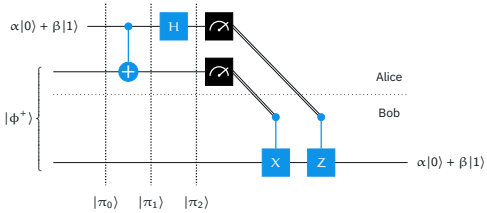
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Teleportation analysis

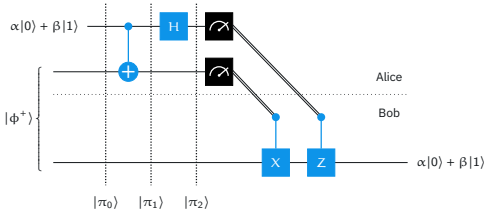


$$|\pi_0\rangle = |\phi^+\rangle \otimes [\alpha|0\rangle + \beta|1\rangle]$$

$$= \left[\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle\right] \otimes [\alpha|0\rangle + \beta|1\rangle]$$

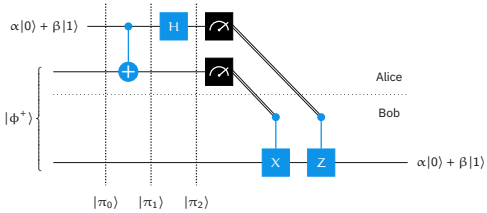
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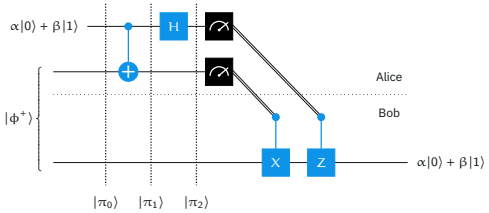
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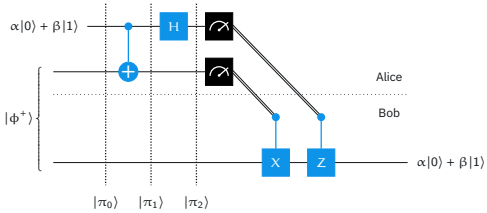


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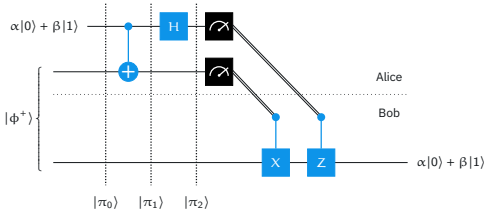
$$|\pi_2\rangle = \frac{\alpha|00\rangle|+\rangle + \alpha|11\rangle|+\rangle + \beta|01\rangle|-\rangle + \beta|10\rangle|-\rangle}{\sqrt{2}}$$

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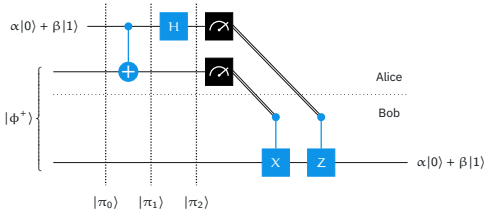
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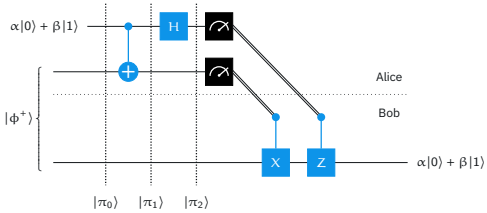
$$= \frac{\alpha|00\rangle(|0\rangle + |1\rangle) + \alpha|11\rangle(|0\rangle + |1\rangle) + \beta|01\rangle(|0\rangle - |1\rangle) + \beta|10\rangle(|0\rangle - |1\rangle)}{2}$$

Teleportation analysis



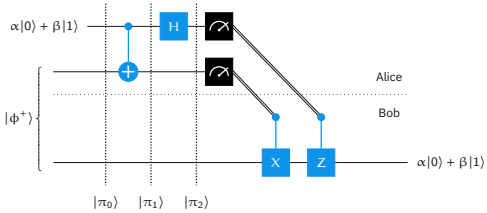
$$
\begin{aligned}
|\pi_2\rangle &= \frac{\alpha|00\rangle|+\rangle + \alpha|11\rangle|+\rangle + \beta|01\rangle|-\rangle + \beta|10\rangle|-\rangle}{\sqrt{2}} \\
&= \frac{\alpha|00\rangle(|0\rangle + |1\rangle) + \alpha|11\rangle(|0\rangle + |1\rangle) + \beta|01\rangle(|0\rangle - |1\rangle) + \beta|10\rangle(|0\rangle - |1\rangle)}{2} \\
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\end{aligned}
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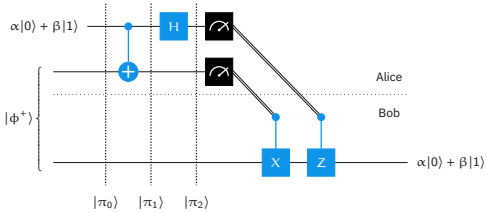
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&= \frac{\alpha|000\rangle + \alpha|001\rangle + \alpha|110\rangle + \alpha|111\rangle + \beta|010\rangle - \beta|011\rangle + \beta|100\rangle - \beta|101\rangle}{2} \\
&= \frac{1}{2}(\alpha|0\rangle + \beta|1\rangle)|00\rangle + \frac{1}{2}(\alpha|0\rangle - \beta|1\rangle)|01\rangle + \frac{1}{2}(\alpha|1\rangle + \beta|0\rangle)|10\rangle + \frac{1}{2}(\alpha|1\rangle - \beta|0\rangle)|11\rangle
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Teleportation analysis



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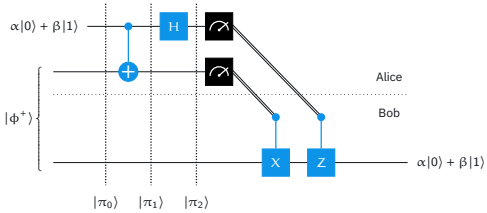
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$$\Pr(ab = 00) = \frac{1}{4}\left\|\alpha|0\rangle + \beta|1\rangle\right\|^2 = \frac{1}{4}$$

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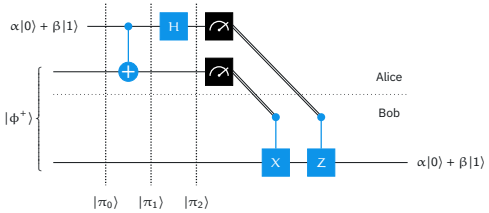
$$\Pr(ab = 00) = \frac{1}{4}\|\alpha|0\rangle + \beta|1\rangle\|^2 = \frac{1}{4}$$

$$\Pr(ab = 01) = \frac{1}{4}\|\alpha|0\rangle - \beta|1\rangle\|^2 = \frac{1}{4}$$

$$\Pr(ab = 10) = \frac{1}{4}\|\alpha|1\rangle + \beta|0\rangle\|^2 = \frac{1}{4}$$

$$\Pr(ab = 11) = \frac{1}{4}\|\alpha|1\rangle - \beta|0\rangle\|^2 = \frac{1}{4}$$

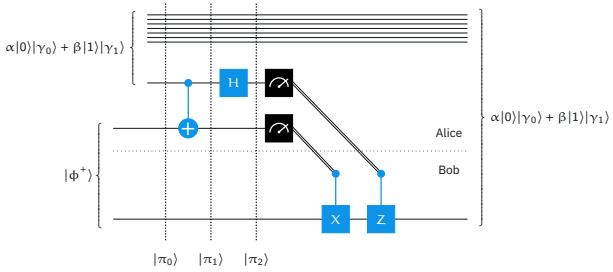
Teleportation analysis



$$|\pi_2\rangle = \frac{1}{2}\left(\alpha|0\rangle + \beta|1\rangle\right)|00\rangle + \frac{1}{2}\left(\alpha|0\rangle - \beta|1\rangle\right)|01\rangle + \frac{1}{2}\left(\alpha|1\rangle + \beta|0\rangle\right)|10\rangle + \frac{1}{2}\left(\alpha|1\rangle - \beta|0\rangle\right)|11\rangle$$

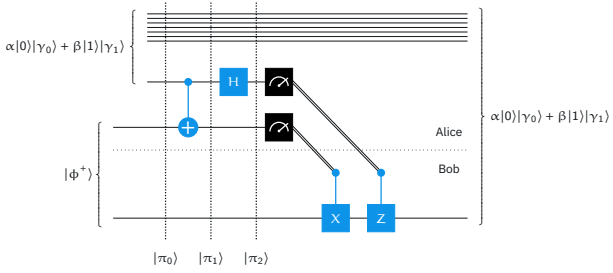
a b	Probability	Conditional state of (B, A, Q)	Operation on B	Final state of B
00	$\frac{1}{4}$	$\left(\alpha\|0\rangle + \beta\|1\rangle\right)\|00\rangle$	$\mathbb{1}$	$\alpha\|0\rangle + \beta\|1\rangle$
01	$\frac{1}{4}$	$\left(\alpha\|0\rangle - \beta\|1\rangle\right)\|01\rangle$	Z	$\alpha\|0\rangle + \beta\|1\rangle$
10	$\frac{1}{4}$	$\left(\alpha\|1\rangle + \beta\|0\rangle\right)\|10\rangle$	X	$\alpha\|0\rangle + \beta\|1\rangle$
11	$\frac{1}{4}$	$\left(\alpha\|1\rangle - \beta\|0\rangle\right)\|11\rangle$	ZX	$\alpha\|0\rangle + \beta\|1\rangle$

Teleportation analysis



$$|\pi_0\rangle = \frac{1}{\sqrt{2}}\left(\alpha|00\rangle|0\rangle|\gamma_0\rangle + \alpha|11\rangle|0\rangle|\gamma_0\rangle + \beta|00\rangle|1\rangle|\gamma_1\rangle + \beta|11\rangle|1\rangle|\gamma_1\rangle\right)$$

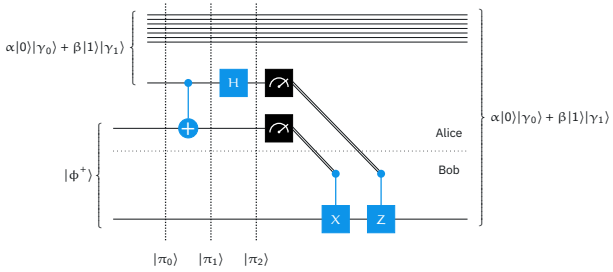
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$$|\pi_1\rangle = \frac{1}{\sqrt{2}}\Big(\alpha|00\rangle|0\rangle|\gamma_0\rangle + \alpha|11\rangle|0\rangle|\gamma_0\rangle + \beta|01\rangle|1\rangle|\gamma_1\rangle + \beta|10\rangle|1\rangle|\gamma_1\rangle\Big)$$

Teleportation analysis

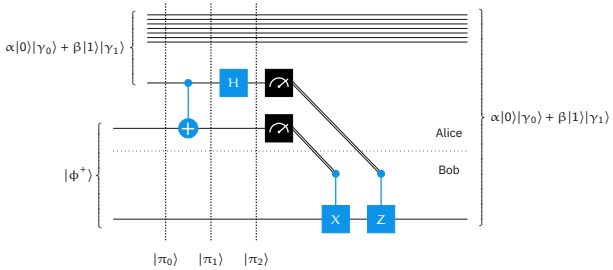


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$$|\pi_2\rangle = \frac{1}{\sqrt{2}}\Big(\alpha|00\rangle|+\rangle|\gamma_0\rangle + \alpha|11\rangle|+\rangle|\gamma_0\rangle + \beta|01\rangle|-\rangle|\gamma_1\rangle + \beta|10\rangle|-\rangle|\gamma_1\rangle\Big)$$

Teleportation analysis



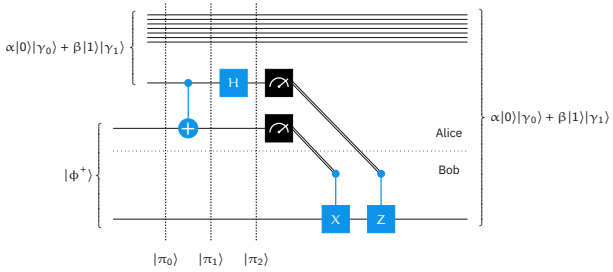
$$|\pi_0\rangle = \frac{1}{\sqrt{2}}\Big(\alpha|00\rangle|0\rangle|\gamma_0\rangle + \alpha|11\rangle|0\rangle|\gamma_0\rangle + \beta|00\rangle|1\rangle|\gamma_1\rangle + \beta|11\rangle|1\rangle|\gamma_1\rangle\Big)$$

$$|\pi_1\rangle = \frac{1}{\sqrt{2}}\Big(\alpha|00\rangle|0\rangle|\gamma_0\rangle + \alpha|11\rangle|0\rangle|\gamma_0\rangle + \beta|01\rangle|1\rangle|\gamma_1\rangle + \beta|10\rangle|1\rangle|\gamma_1\rangle\Big)$$

$$|\pi_2\rangle = \frac{1}{\sqrt{2}}\Big(\alpha|00\rangle|+\rangle|\gamma_0\rangle + \alpha|11\rangle|+\rangle|\gamma_0\rangle + \beta|01\rangle|-\rangle|\gamma_1\rangle + \beta|10\rangle|-\rangle|\gamma_1\rangle\Big)$$

$$= \frac{1}{2}\Big(\alpha|0\rangle|00\rangle|\gamma_0\rangle + \alpha|0\rangle|01\rangle|\gamma_0\rangle + \alpha|1\rangle|10\rangle|\gamma_0\rangle + \alpha|1\rangle|11\rangle|\gamma_0\rangle$$
$$+\beta|1\rangle|00\rangle|\gamma_1\rangle - \beta|1\rangle|01\rangle|\gamma_1\rangle + \beta|0\rangle|10\rangle|\gamma_1\rangle - \beta|0\rangle|11\rangle|\gamma_1\rangle\Big)$$

Teleportation analysis



$$|\pi_2\rangle = \frac{1}{2}\Big(\alpha|0\rangle|00\rangle|\gamma_0\rangle + \alpha|0\rangle|01\rangle|\gamma_0\rangle + \alpha|1\rangle|10\rangle|\gamma_0\rangle + \alpha|1\rangle|11\rangle|\gamma_0\rangle + \beta|1\rangle|00\rangle|\gamma_1\rangle - \beta|1\rangle|01\rangle|\gamma_1\rangle + \beta|0\rangle|10\rangle|\gamma_1\rangle - \beta|0\rangle|11\rangle|\gamma_1\rangle\Big)$$

ab	Probability	Conditional state of (B, R, A, Q)	Operation on B	Final state of (B, R)
00	$\frac{1}{4}$	$(\alpha\|0\rangle\|\gamma_0\rangle + \beta\|1\rangle\|\gamma_1\rangle)\|00\rangle$	$\mathbb{1}$	$\alpha\|0\rangle\|\gamma_0\rangle + \beta\|1\rangle\|\gamma_1\rangle$
01	$\frac{1}{4}$	$(\alpha\|0\rangle\|\gamma_0\rangle - \beta\|1\rangle\|\gamma_1\rangle)\|01\rangle$	Z	$\alpha\|0\rangle\|\gamma_0\rangle + \beta\|1\rangle\|\gamma_1\rangle$
10	$\frac{1}{4}$	$(\alpha\|1\rangle\|\gamma_0\rangle + \beta\|0\rangle\|\gamma_1\rangle)\|10\rangle$	X	$\alpha\|0\rangle\|\gamma_0\rangle + \beta\|1\rangle\|\gamma_1\rangle$
11	$\frac{1}{4}$	$(\alpha\|1\rangle\|\gamma_0\rangle - \beta\|0\rangle\|\gamma_1\rangle)\|11\rangle$	ZX	$\alpha\|0\rangle\|\gamma_0\rangle + \beta\|1\rangle\|\gamma_1\rangle$

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- Teleportation is not an application of quantum information — it's a way to perform *quantum communication.*

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- Teleportation is not an application of quantum information — it's a way to perform *quantum communication.*
- Teleportation motivates *entanglement distillation* as a means to reliable quantum communication.
- Beyond its potential for communication, teleportation also has fundamental importance in the study of quantum information and computation.

What we learned

1

Entanglement. A shared entangled pair is the resource.

2

Measurement and a call. Alice's two outcomes tell Bob which gates to apply.

3

No faster-than-light signal. Bob needs Alice's classical bits.

Adapted from IBM Qiskit in Classrooms: Quantum teleportation

Now, open the teleportation notebook

